星間物理学

IRIS '95
ASTRO-F '03

(1) Outline
(2) HII 領域（電離水素）
(3) HI 領域（中性水素）
(4) 星間ダスト
(5) 分子雲
(6) 星形成・惑星形成
(7) 光解離領域
(8) 銀河中心
(9) 銀河系、他の銀河との比較
(10) Starburst Galaxies
(11) 活動銀河核（AGN）
(12) 赤外線銀河
(13) 銀河団

$H\alpha$ $n=3\to2$

(1) Outline

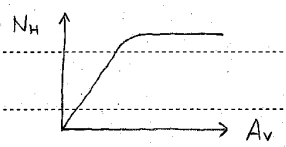
1気圧、常温 $n = 3\times10^{19}\ cm^{-3}$ H 21cm $A = 10^{-15}\ s$

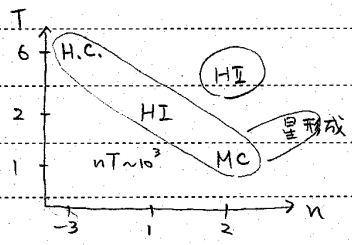
三体反応 ×　　H_2 : $A = 10^{-5}\ s$

M16、HST 星、赤くなっている ← ダスト 0.1μm

0.5μm	10μm	100μm	1mm
10^4 K	510k	51k	5k
2.4eV	0.12eV	0.01eV	1meV

	n	T
電離領域	$1\sim10^3\ cm^{-3}$	8000 ~ 10,000 K
中性 〃	1~10	100
分子雲 〃	$10^{2\sim5}$	10 ~ 50
Hot Coronal Gas	10^{-3}	10^6





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エネルギー

Stellar Radiation →
SN Kinetic → (ISM) → Radiation ⇒ obs
磁場 →

化学進化　Primodial Gas H, He ⟶ Contaminated Gas
↑ 星 mass loss　↑ SN
He, C, N, O, Fe

(2) HII 領域　水素が電離　可視光 | 水素電離再結合線 $A=10^{4-8}\,s^{-1}$ ▷ Apr 18

M42 トラペジウム　O6 - O7 | $n=3\to2$ (H_α)　$4\to2$ (H_β) Balmer Line

bright bar | O^+, O^{++}, S^+ ⇒ Forbidden Lines $A=1\text{-}10^{-5}\,s^{-1}$

5007Å
4996Å

$n_O/n_H = 6\times10^{-4}$

$T_e = 5000\,K \sim 10^4\,K$

$n_C/n_H = 3\times10^{-4}$

Source of (Ionization ⇒ Early Type Stars $T_* = 2\times10^4 - 6\times10^4\,K$
Heating

Ionization Potential

For simplicity ⇒ H のみ　$E_H = 13.6\,eV$ (912 Å)

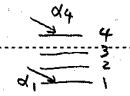
Photoionization　$h\nu > E_H$　$E_K = h\nu - E_H$

Recombination　$H + h\nu \rightleftharpoons p + e$

ある単位体積

Recombination Rate $\propto n_p n_e = n_e^2$

比例係数 $\alpha_n(T)$
↑ n 準位



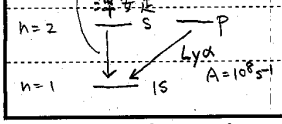
電子の運動エネルギー

Recombine Photon Energy　$h\nu = \frac{E_H}{n^2} + kT_e$

$= E_H\left(1/n^2 + 0.1\,(T_e/10^4K)\right)$

$n=1$　$h\nu = 1.1\,E_H$
$n=2$　$0.35\,E_H$

Two Photon Decay



中性水素 $n=1$?

n に Recombine → n-1, n-2 --- Radiative Transition　$A=10^{4-8}\,s^{-1}$

一方 Ionization Rate $10^{-8}\,s^{-1}$

COLLEGE NOTES / n090

Effective Recombination Rate $\sum_{n=2}^{\infty} \alpha_n(T)\, n_e^2 = \alpha_B(T)\, n_e^2$

Ionization Degree : x $\quad n_e = x n$

$n_H = (1-x) n$

Ionization Rate $\quad \sigma_0$ Ionization cross section ($7\times10^{-18}\,cm^2$)

J_{Lyc} : Ionization Continuum Photon Numbers

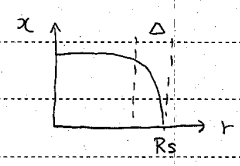
$\sigma_0 n_H J_{Lyc} = \sigma_0 (1-x) n J_{Lyc}$

" $\frac{N_{Lyc}}{4\pi r^2}$ $\quad N_{Lyc} = \int_{h\nu > E_H} \frac{L_{Lyc}}{h\nu} d\nu$

$x^2 n^2 \alpha_B(T_e) = \sigma_0 (1-x) n J_{Lyc} \rightarrow \frac{x^2}{1-x} = \frac{J_{Lyc}}{n} \frac{\sigma_0}{\alpha_B(T_e)}$

$n = 10^2$

$r = 1pc = 3\times10^{18} cm$



↓

$1-x = 3.4\times10^{-5}$

$x \simeq 1$ Almost Fully Ionized

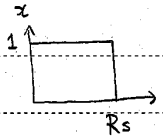
O6.5 $N_{Lyc} = 10^{49} s^{-1}$

$J_{Lyc} = 8.4\times10^{10} cm^{-2} s^{-1}$

mean free path $\Delta = \frac{1}{\sigma_0 n_H} \sim 4.8\times10^{-2} n^{-1}$ pc

$\ll R_s$ Edge は Sharp

仮定



$N_{Lyc} = \alpha_B n_e^2 \frac{4\pi}{3} R_s^3 \rightarrow R_s = \left(\frac{3}{4\pi} \frac{N_{Lyc}}{n_e^2 \alpha_B}\right)^{1/3}$ strömgren Sphere

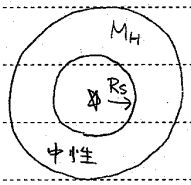
$= 3$ pc

	T_*	$\log N_{Lyc}$	R_s ($n=1$)	$n=10^2$
O5	48000	49.67	108 pc	5 pc
O9	32000	48.24	34 pc	1.6 pc
B0.5	26200	46.83	12 pc	0.6 pc

Evolution of HII regions : 密度 ↘

$n_H \rightarrow n_e + n_p$ 2倍) 圧力大 (×200)

$T = 10^2 K \rightarrow 10^4 K$



単位面積あたり

$\frac{M}{4\pi R_s^2}$

$\Delta P = P_{HII} = 2 n_e k T_e$

$\frac{M_H}{4\pi R^2} \ddot{R} = 2 n k T_e$

$\dot{R} = \left(\frac{4\pi n k T_e}{3M} R^3\right)^{-1/2}$

$\tau = \frac{R_s}{\dot{R}_s} = \left(\frac{3M}{4\pi n_e k T} \frac{1}{R^3}\right)^{1/2}$

$M = 1 M_\odot$

$n = 10^4$, $T = 10^4$

$R_s = 0.03$ pc

$\dot{R} = 3$ km/s

$\tau = 1.5\times10^4$ yr

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星が生まれると Compact HII Region $n=10^4$, $r=0.03\text{pc}\sim0.1\text{pc}$ 膨張 →

$v_{IF} > c_s$ Shock Wave ▷Ap 26

ρ_1, T_1 | ρ_0, T_0 ← v_1 | ← v_0

(1) mass flux $I \equiv \rho_0 v_0 = \rho_1 v_1$ ··· Ⓐ

(2) momentum flux $\rho_0 v_0 v_0 - \rho_1 v_1 v_1 = -P_0 + P_1$

$\xi = P_0 + \rho_0 v_0^2 = P_1 + \rho_1 v_1^2$ ··· Ⓑ

(3) energy flux

(i) kinetic energy (per mass) $v^2/2$

(ii) Internal energy $U = c_v T = \frac{1}{\gamma-1}\frac{P}{\rho} = \frac{3}{2}\frac{P}{\rho}$ $(\gamma = \frac{5}{3})$

(iii) Work / time / area $= Pv$

Work / mass $= P/\rho$ $5/2$

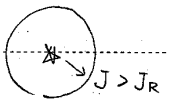
$\xi = E_{kin} + U + W = \frac{v^2}{2} + \frac{1}{\gamma-1}\frac{P}{\rho} + \frac{P}{\rho} = \frac{v^2}{2} + \frac{\gamma}{\gamma-1}\frac{P}{\rho}$

$h\nu + H \rightarrow H^+ + e^-$ $\quad Q + \left(\frac{v_0^2}{2} + \frac{5}{2}\frac{P_0}{\rho_0}\right) = \frac{v_1^2}{2} + \frac{5}{2}\frac{P_1}{\rho_1}$ ··· Ⓒ

$E_{kin} = E_{rad} - E_{ion}$
$\quad\quad\quad

Evolution of HII Region

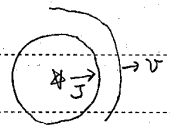
i) J が大きい (Ionizing Flux が大きい) $J \geq J_R$ ⇒ Ionization が急速に進む $v > v_R$



Ionization Front = Shock Front

ii) $J = J_R$

iii) $J_R > J > J_D$ Ionization front ≠ Shock front



iv) $J \leq J_D$ $(u \gg c_s)$

$v \leq v_D = \frac{1}{3}\left(-2u + \sqrt{4u^2 + 9c_s^2}\right) \sim \frac{3c_s^2}{4u} < c_s$ Shock できない, I-front だけ

e.g.) $n_H = 6.4\ cm^{-3}$

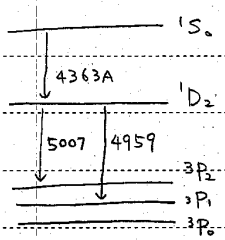
O7

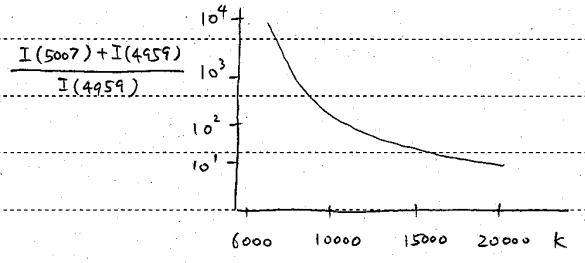
i) $v > v_R$ $\Delta t \sim 8 \times 10^4$ yr

ii) $v_R > v > v_D$ $\Delta t > 2 \times 10^6$ yr

温度を測る

[OIII]





NGC1976 (Ori) T = 8500 ~ 9300 K

M8 8100 K

Heating: Ionization $h\nu - E_{ion} = E_k$

Heating rate $G = \dot{n}_{ion} E_k$ $\dot{n}_{ion} = \dot{n}_R$

Recombination $\mathcal{L} = \frac{3}{2} k T_e \dot{n}_R$

$\mathcal{L} = G$ $T_* = 2 \times 10^4\ K \sim 6 \times 10^4\ K$

$T_{gas} = 1.5 \times 10^4 \sim 4 \times 10^4\ K$ ← 観測と合わない

↑ metal の cooling だった

Lyα 10eV

電離度による

photo ionization → $x_e \sim 1$

collision } CR → $x_e \sim 0.1$

↓ Lyα

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HI 領域　21cm Hyperfine Structure Line

1951 first detect (USA)　$A = 2.87 \times 10^{-15}\ s^{-1}$

van der Hulst (1944)

F=1 ↑ ↑ (p e)　　F=0 ↑ ↓ (p e)

$g = 2F+1 = 3$　　1

A_{10} ↓　↑ C_{01} ↓ C_{10}

F=1

F=0

$\nu = 1{,}420\ GHz$　　$E = 6 \times 10^{-6}\ eV$

$\lambda = 21.106\ cm$　　$T = 0.07\ K$

$C \simeq n_H \sigma v_T$　　$\sigma \sim 10^{-16}\ cm^2$, $v_T \sim 1\ km/s \left(\frac{T}{100K}\right)^{1/2}$, $n_H = 10\ cm^{-3}$

$\sim 1.5 \times 10^{-10}\ s^{-1} \gg A$

Collision によって population が決まる → Boltzmann 分布　$\frac{n_1}{n_0} = \frac{g_1}{g_0} \exp\left(-\frac{\Delta E}{kT}\right) \sim 3$　（$g_1 = 3$）

吸収係数　$K_\nu \equiv -\frac{1}{I_\nu}\frac{dI_\nu}{ds}\ cm^{-1}$

放射係数　ε_ν　$erg\ s^{-1}\ cm^{-3}\ sr^{-1}$

radiative transfer　$dI_\nu = \varepsilon_\nu ds - K_\nu I_\nu ds$　　$S_\nu = \frac{\varepsilon_\nu}{K_\nu}$ Source function

$\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu$　　$\tau_\nu = \int_0^s K_\nu ds$ optical depth

$\rightarrow I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} e^{-\tau_\nu'} S_\nu d\tau_\nu'$

$I_\nu(0) = 0$, 一様　→　$= S_\nu (1 - e^{-\tau_\nu}) \rightarrow S_\nu$　$(\tau_\nu \gg 1)$ thick

$\tau_\nu S_\nu$　$(\tau_\nu \ll 1)$ thin

Source function, thermal → Planck Function　$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(\frac{h\nu}{kT}) - 1}$

$kT \gg h\nu$ (Rayleigh-Jeans)　$\rightarrow \frac{2kT}{c^2}\nu^2$　（T に比例）

T_B Brightness Temperature　$T_B \equiv \left(\frac{c^2}{2k\nu^2}\right) I_\nu = T_K (1 - e^{-\tau_\nu}) = \begin{cases} T_K & \text{thick} \\ \tau_\nu T_K & \text{thin} \end{cases}$

$K_\nu = \frac{1}{4\pi}(n_0 B_{01} - n_1 B_{10}) h\nu f_\nu$　　A_{10} 自発遷移 (s^{-1})

$\varepsilon_\nu = \frac{1}{4\pi} n_1 A_{10} h\nu f_\nu$　　B_{10} stimulated emittion $B_{01} I_\nu\ (s^{-1})$

f_ν line shape function　$\int_{-\infty}^{\infty} f_\nu d\nu = 1$　　B_{01} 吸収 $B_{10} I_\nu\ (s^{-1})$

$g_0 B_{01} = g_1 B_{10}$

$\rightarrow K_\nu = \frac{c^2}{8\pi\nu^3} n_0 \left(1 - e^{-\frac{h\nu}{kT}}\right) \frac{g_2}{g_1} A_{21} f_\nu$　　$A_{10} = \frac{2h\nu^3}{c^2} B_{10}$

$\simeq \frac{c^2 A_{10}}{8\pi\nu^2} \frac{3h n_H}{4kT} f_\nu = 2.6 \times 10^{-15} \frac{n_H}{T} \frac{1}{\Delta v}$　$\left(f_\nu = \frac{1}{\Delta v}\right)$

$N_H = n_H L = (2.6 \times 10^{-15})^{-1} K_\nu L T \Delta v = 3.8 \times 10^{14} T_B \Delta v = 1.8 \times 10^{18} T_B \Delta v\ cm^{-2}$

Δv: km/s

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注1)

continuum T_c → (T_K, τ_ν) → ▷

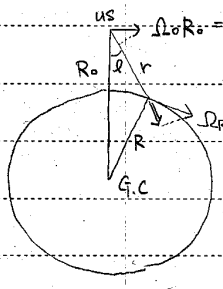
$$T_B = T_c e^{-\tau_\nu} + T_K (1 - e^{-\tau_\nu})$$

$$\Delta T_B = T_B - T_c = (T_K - T_c)(1 - e^{-\tau})$$

e.g.) G.C Sgr A Continuum Source ⇒ Absorbtion

注2) thin ? $T_\nu = 5.6 \times 10^{-19} \frac{N_H}{T} \frac{1}{\Delta v} = \left(\frac{N_H}{1.8\times10^{18}\,cm^{-2}}\right)\left(\frac{100K}{T}\right)\left(\frac{10km/s}{\delta v}\right)$

G.C.

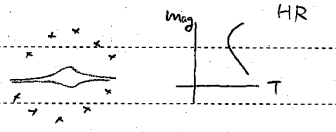


視線速度 $v = \Omega R \cos(90° - \ell - \theta) - \Omega_0 R_0 \cos(90° - \ell)$

$= R_0(\Omega(R) - \Omega_0)\sin\ell$

① R_0 の決定

Globular cluster の分布



$R_0 = 8.5\,kpc \pm 1\,pc$

② R_0, Ω_0 の決定

動かない
動きが分かっている
↑
Globular Clusters
Local Group of Galaxies

} 天体に対して Solar System の動きを測る (Local Standard of Rest)

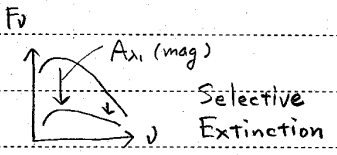
$R_0\Omega_0 = 220\,km/s \pm 20\,km/s$

Interstellar Dust

May 17

観測的事実　1) dark cloud の存在

2) 〃　では星は赤くなる



$Q_{ext} \propto \lambda^{-1}$ (optical)

$a \simeq \lambda$

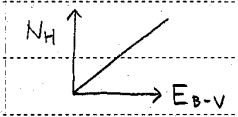
A_λ (mag) : Absolute Extinction

⇒ 観測的に求めることは困難

$E(\lambda_1 \lambda_2) \equiv A_{\lambda_1} - A_{\lambda_2}$ ⇐ 求めることが容易 ⇐ 星の本来の色と obs. color との比較

(相対的)

E_{B-V} : 4350 Å – 5550 Å



$\frac{N_H}{E_{B-V}} = 5.9 \times 10^{21}\ cm^{-2}/mag$

$R = \left[\frac{E_{V-\lambda}}{E_{B-V}}\right]_{\lambda 3000} = \frac{A_V}{E_{B-V}} \simeq 3$ (≃5 くらいの時もある)　← Extinction curve の傾き

$\frac{N_H}{A_V} = 2 \times 10^{21}\ cm^{-2}\ mag^{-1}$　　$\frac{A_V}{L} = 1.8\ mag\ kpc^{-1}$

Extinction の源は？

1) free electron → Thomson 散乱　$\sigma_T = 6 \times 10^{-25}\ cm^{-2}$　→ $n_e = 45\ cm^{-3}$

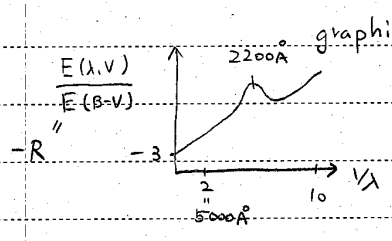
↑ 波長によらない

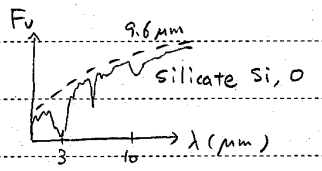
2) 大きな粒子　$a \gg \lambda$　σが波長によらない

3) 非常に小さな粒子　$a \ll \lambda$ → Reyleigh 散乱 $\sigma \propto \lambda^{-4}$

4) $a \sim \lambda$ くらいの粒子

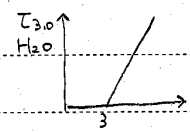
粒子の成分





Volatile Molecules

H_2O	100K
CO_2	40-50K
CO	20K

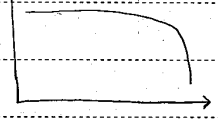


off set　Avの小さい所、UVで こわされる

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より重い元素

D(x) ↑ ---------Cosmic Abundance　　⇒ 重い元素のかなりは dust



UV, opt) Abs

ガス内に存在している元素の量が Cosmic Abundance からどれだけ減っているか

Dust の光学的性質

Q_{ext}

$\sigma_{ext} = Q_{ext} \cdot \pi a^2$　　albedo : $\gamma = \frac{Q_{scatter}}{Q_{ext}}$

$Q_{ext} = Q_{scatter} + Q_{abs}$

Refractive Index　$m = n - ik$　(k: absorption)　　$k=0 \rightarrow \gamma=1$, $n=0 \rightarrow \gamma=0$

$x \equiv 2\pi a/\lambda$

$x \ll 1$ Rayleigh Scattering

$x \sim 1$ Mie Scattering

Spherical, $\lambda \gg a$ (Rayleigh)

$Q_{scatter} \simeq \frac{8}{3} x^4 \, \mathrm{Real}\left(\frac{m^2-1}{m^2+2}\right)^2$　　van de Hulst

$Q_{abs} \simeq 4x \, \mathrm{Imag}\left(\frac{m^2-1}{m^2+2}\right)$　　"Light Scattering by Small Particles"

($x \propto \lambda^{-1}$)　　John Wile & Sons

Dust の存在量

$A_V/L = 1.8$ mag kpc^{-1}

ダストの空間密度 ρ_d (g cm^{-3})　　$\rho_d = \frac{N_d m_d}{L} = \frac{N_d}{L}\frac{4\pi}{3}a^3 s$　(N_d: ダストの column density)

ダスト粒子 1コ　$m_d = \frac{4\pi}{3} a^3 s$　　$A_V = 1.086 \, N_d \, Q_{ext} \, \pi a^2$

(a: 0.1 μm, s: 物質の密度 ~3, Q_{ext}: 0.6)

$\rightarrow \rho_d = 1.8 \times 10^{-26}$ g cm^{-3}

$\rho_{dust}/\rho_{gas} = 1/160$　　$N_H = 2.1 \times 10^{21}$ cm^{-2} A_V

質量比 Heよりも重い元素の割合 1.4(-2)　　$n_H = 1.2$ cm^{-3}

C, N, O のかなりは dust　　$\rho_{gas} = 2.2 \times 10^{-24}$ g cm^{-3}

COLLEGE NOTES / n090　　Source p.9 / 9 of 18 selected pages

IRAS 12, 25, 60, 100 μm Infrared Cirrus May 24

ダストの温度

$W_{abs} = \int \frac{L_\lambda}{4\pi r^2} Q_{abs} \pi a^2 d\lambda$ $\frac{L_\lambda}{4\pi r^2} = \left(\frac{R_*^2}{r^2}\right) \pi B_\lambda(T_*)$ w dilution factor

Local Interstellar Radiation Field $T_* = 10^4$ K

$W = 10^{-14}$

$W_{em} = \int Q_{em} 4\pi a^2 B_\lambda(T_d) d\lambda$

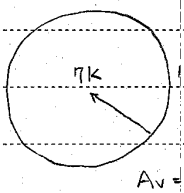
仮定 $Q_{abs}(\lambda) = Q_{em}(\lambda) = 1$ ⇒ $T_d = 3.1$ K

1つの1

$\hat{T}_d$ (obs) = 20K

各々の波長で $Q_{abs} = Q_{em}$?

1つの2 abs : UV ~ Blue $Q_{abs}(UV) \sim 1$



em : FIR $Q_{em}(FIR) \propto \lambda^{-\beta}$ $\beta = 1-2$

→ $T_d = 15K - 20K$

$A_V = 10$ mag $F_\lambda = N_d \pi a^2 Q_\lambda B_\lambda(T_d) \propto \lambda^{-\beta} B_\lambda(T_d)$

$N_H = 2 \times 10^{22}$ cm^{-2}

長い波長が早く落ちる

$\lambda_{peak} = 3000 \left(\frac{5}{\beta+5}\right) T_d^{-1}$ Wien's displacement law

gas/dust ratio

$\rho_{gas} / \rho_{dust} = 160$ (Extinction)

$I_\lambda = \tau_\lambda B_\lambda(T_d)$ $K_{100} = 41$ cm^2 g^{-1} $\rho_{dust} = I_\lambda / B_\lambda(T_d) K_\lambda L$

$\tau_\lambda = \rho_{dust} K_\lambda L$ $\rho_H = N_H / L \cdot m_H$

IRAS / 21cm obs → $\frac{\tau_{100}}{N_H} = 3 \times 10^{-25}$ cm^2/nuclear $\frac{\rho_{gas}}{\rho_{dust}} = 4.3 \times 10^{26} \frac{\tau_\lambda}{N_H K_\lambda} = 330$ (Emission)

Emission に基く Estimate は 大きめ

問題 1) K_λ の不定性

2) Relatively Warm Dust のみ Cold の存在

短い波長の超過は?

今まで $W_{abs} = W_{em}$ これは常に本当?

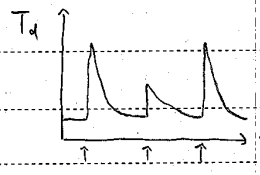
Very Small Grain Heat Capacitance $C_V = 2 \times 10^{-7}$ erg K^{-1} ($a = 0.1 \mu m$ = 1000Å) classical grains

9×10^{-14} $a = 8$Å very small grains

$h\nu = 10$ eV 1コ当たる

$\Delta T = h\nu / C_V = 230$K (VSG)

10^{-4} K (CG)



Stochastic Heating

$L = \sigma T^4 4\pi a^2 Q_{abs} = 3 \times 10^{-12}$ erg s^{-1} → $\Delta t = \frac{h\nu}{L} = 6$ sec

(Q = 1)

COLLEGE NOTES / n090 Source p.10 / 10 of 18 selected pages

1) Spectrum が Input Radiation の Intensity によらず Spectrum による

2) $T_{obs} \gg T_{eq}$ を実現可能

Unidentified IR Bands　Vibrational Transition　3.3 μm　C-H streching

Polycyclic Aromatic Hydrocarbons (PAHs)　6.2　C-C　7.7　C-C　Streching　8.3　C-H　11.3　C-H　bending　in plane　out of plane

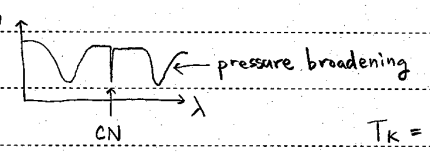
$h\nu = (3N-6)\int_0^{T_p} C(T)\,dT$

↑ mode の数　↳ 一つの vibrational mode あたり

$N = 60$ atom, $h\nu = 10$ eV → $T_p = 1000$ K

分子雲　May 31

1940 3 Oph Optical　O-type



$T_K = 3$ K

Radio obs.

1963 OH $\lambda = 18$ cm

1968 NH_3 $\lambda = 1.3$ cm) towns

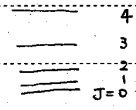
1969 H_2O $\nu = 22$ GHz ← メーザー (Teff < 0, Highly excited, Stimulated emission)

1969 H_2CO ホルムアルデヒド

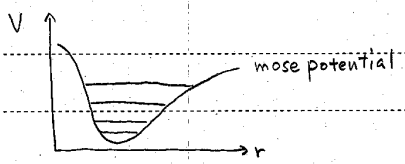
1970 CO $\lambda = 2.6$ mm $\nu = 115$ GHz Penzios, Wilson (Bell-Lab)

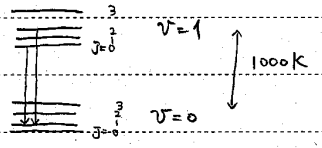
観測

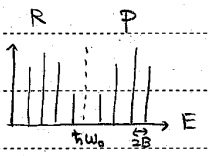
回転 $E_J = hBJ(J+1)$　$\Delta J = \pm 1$



Vibrational (-Rotational) Transition　$E(v) = \hbar\omega_0(v+\frac{1}{2})$







$v=1$ $J=2$ → $v=0$ $J=1$

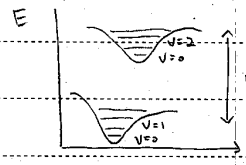
$v=1 \to 0$ $R(1)$ ← おちた先の J　$\Delta v = 1$

$v=1$ $J=0$ → $v=0$, $J=1$　$\Delta E = \hbar\omega_0 + 2(J+1)B$　R-branch

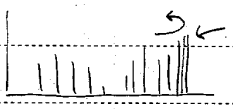
$P(1)$　$= \hbar\omega_0 - 2JB$　P-branch

COLLEGE NOTES / n090　Source p.11 / 11 of 18 selected pages

Electronic Transition



Electric State が異なる → B, ω₀ が異なる



← Band Head ができることが多い

Opt ~ UV absorption

観測から物理量へ

CO (115GHz) 安定 E ~ 11eV 豊富 critical density 低い $10^3\ cm^{-3}$

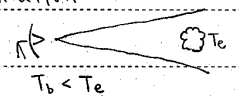
$^{12}C^{16}O$ $\tau \gg 1$

$^{13}C^{16}O$ Optically thin $\quad T_b = T_e(1-e^{-\tau}) = \begin{cases} T_e & \tau \gg 1 \quad ^{12}CO \\ \tau T_e & \tau \ll 1 \quad ^{13}CO \end{cases}$

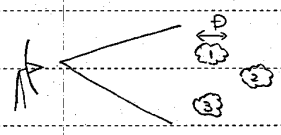
$n(^{12}CO)/n(^{13}CO) \simeq 90$ (Solar System)

40 (ISM)

dilution



$^{12}CO \rightarrow N_{H_2}$ Conversion Factor



$I_{CO} \propto \sum_i T_R^i A^i \sigma^i$ (A: 立体角, σ: 速度幅)

$I_{CO} \propto N \langle T_R \rangle \langle A \rangle \langle \sigma \rangle$ [K·km/s]

K km/s

各々の cloud → 重力的に bound (Virial $\sigma \propto (M/D)^{1/2}$

$M = \frac{4}{3}\pi \left(\frac{D}{2}\right)^3 \mu \bar{n}$

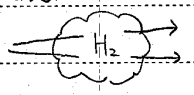
$N_{H_2} \propto N \bar{n} D^3$ $\quad \rightarrow \sigma \propto (\mu \bar{n})^{1/2} D$

$X = \frac{N_{H_2}}{I_{CO}} = \frac{2}{3} \frac{\bar{n}\langle D \rangle}{(2\pi)^{1/2} \langle T_R \rangle \langle \sigma_r \rangle} = \frac{1}{\pi \langle T_R \rangle} \left(\frac{2\bar{n}}{G\mu}\right)^{1/2}$

$\bar{n} = 300\ cm^{-3}$, $T_R = 6\ K$ → $= 2.5 \times 10^{20}\ cm^{-2}/K\ km/s$

水素分子 (H_2 Molecule) ▷ Jun 7

UV 吸収観測

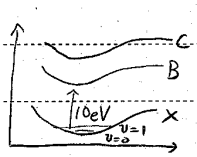
OB type 

us

電子基底状態 X

第一電子励起 B Lyman-Bands

第二電子励起 C Werner-Bands



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四重極放射 Selection rule $\Delta J = +2$ S Branch　　Electronic Transition

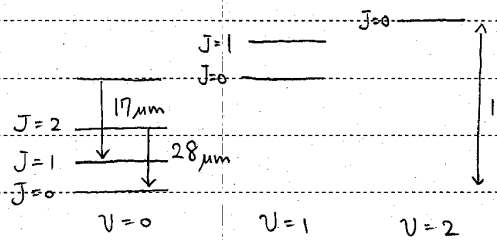
vib-rot transition → 0 Q

-2 O　　Dipole $\Delta J = \pm 1$ P Branch

例 $v = 1 \to 0$ S(0)　　R Branch

$J = 2 \to 0$　　UV absorption → diffuse clouds

Pure Rotational Transition $E_J = hB_0 J(J+1)$　　$\Delta J = \begin{pmatrix} \pm 2 \\ 0 \end{pmatrix}$



$A_{20} = 3 \times 10^{-11}\ s^{-1}$

$J = 2 \to 0$　$\lambda = 28.2\mu m$　510K

$3 \to 1$　$17\mu m$　850K

$A_{31} = 5 \times 10^{-11}\ s^{-1}$

吸収は $\Delta v = 1$ km/s $\tau = 1 \to N_{H_2} = 10^{24}\ cm^{-2}$

Vib-Rotational Transition $E_v = h\nu(v + \frac{1}{2})$　　$\Delta v = 0, \pm 1, \pm 2, \cdots$

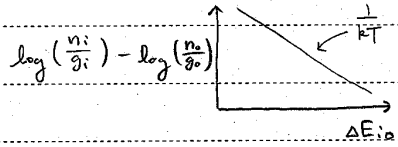
$\Delta J = 0, \pm 2$

$\Delta v = 1$　6000K　$2\mu m$　$A = 10^{-7}\ s^{-1}$

観測 → 励起メカニズム

Level Population Diagram　　Boltzman $\frac{n_2}{n_1} = \frac{g_2}{g_1} \exp\left(-\frac{\Delta E_{12}}{kT}\right)$

$\to \log(n_2/g_2) - \log(n_1/g_1) = -\frac{\Delta E_{12}}{kT}$



Ori-KL　$T \simeq 2000$K

同じ v, 異る J) T_{rot}

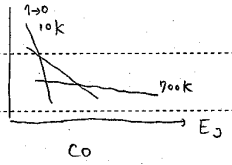
高温 Mol. Gas の存在 → Shock excitation　異る v, 同じ J) T_{vib}

NGC 2023 Ref. Neb

$T_v > 3500$K　　Photo excitation

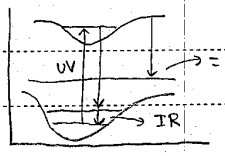
$T_r = 1000 - 1500$K　　しかし直接赤外線での excitation は困難

B係数小



Co

UV absorption



→ こわれる 10%

UV Fluorescence

COLLEGE NOTES / n090　　Source p.13 / 13 of 18 selected pages

$g = g_s g_J$ $\quad g_J = 2J+1$

$g_s = \begin{cases} \text{atomic spin 平行 ortho } H_2 & g_s = 3 \quad J = \text{odd} \ (J = 1, 3, 5, \cdots) \\ \text{反: para} & = 1 \quad J = \text{even} \ (J = 0, 2, 4, \cdots) \end{cases}$

ortho ※ para

NGC2023 $\frac{\text{ortho}}{\text{para}} = 1.8 \pm 3$ $\quad 3 \quad T > 250\,K$

$1.8 \quad T \sim 100\,K$

Fluorescent H_2 が detect された所

ortho/para $= 1 \sim 1.8 \quad T = 80 \sim 110\,K$

c.f. HD $J = 1 \to 0 \quad 112\,\mu m$ $\quad \frac{HD}{H_2} = 10^{-5}$

$2 \to 1 \quad 56\,\mu m$ $\quad$ ISO

H_2 の生成

obs results H_2 の割合 $f = \frac{2n(H_2)}{n(H) + 2n(H_2)}$ $\quad$ UV Absorption

$f \simeq 10^{-6} - 10^{-1}$

平衡 $\beta n(H_2) = R n^2(H)$

$R = \frac{\beta}{\Sigma n} f$ $\quad$ e.g. δ Ori $\begin{cases} n = 10 - 30\,cm^{-3} \\ \beta = 44 \times 10^{-10}\,s^{-1} \\ f = 6 \times 10^{-6} \end{cases} \Rightarrow R = 1.5 \times 10^{-17}\,cm^3 s^{-1}$

C, N, O の H との collision $1\,yr^{-1}$ $\quad (n = 10^2\,cm^{-3})$ $\quad$ ▷Jun 14

C, N, O 同志 $\quad 10^{-4}\,yr^{-1}$

1) neutral - neutral (collision) $\quad k \sim \sigma v \quad 10K$ の $v = 5 \times 10^4\,cm/s \quad \sigma = \pi (1Å)^2 = 3 \times 10^{-16}\,cm^2$

$k \sim 10^{-11}\,cm^3 s^{-1}$

$k \propto \exp(-\frac{E}{T})$ e.g. $\begin{cases} H + O_2 \to OH + O & E = 8500K \\ OH + CO \to H + CO_2 & E = 300K \end{cases}$

2) ion - molecule

実験 $k \sim 10^{-9}$, T-dependence 少 など

σ 分極: $de \propto \frac{\alpha}{r^2}$ $\quad \alpha$: 分極率

$F = \alpha \left\{ \frac{1}{r^2} - \frac{1}{(r+d)^2} \right\} = \alpha \frac{2d}{r^3} \propto \frac{\alpha}{r^5} \quad (d \ll r)$

$b_0 = \left(\frac{4e^2 \alpha}{\mu v^2} \right)^{1/4}$ ← reduced mass

$k = \pi b_0^2 v = 2\pi e \left(\frac{\alpha}{\mu} \right)^{1/2}$

speices	α
H_2	$0.79 \times 10^{-24}\,cm^3$
CO	1.95
H_2O	1.40

COLLEGE NOTES / n090

3) radiative association

滞在時間 $\Delta t \simeq r_{AB}/v = 10^{-13}$ sec | Δt の間に radiation がおきる確率 $P < 10^{-5}$

$A \lesssim 10^8\ s^{-1}$ (Lyα) | k = collision rate × $P \sim 10^{-16}\ cm^3/s$

ion が大要

1) diffuse clouds UV photoionization

2) molecular clouds Cosmic Ray Ionization $H_2 + cr \rightarrow H_2^+ + cr$

$\xi = 10^{-17}\ s^{-1}$ $H_2^+ + H \rightarrow H_3^+$

反応 rate

$k = 10^{-9}$, $n_H = 10^2$, $n_H/n_H = 10^{-4}$ $\rightarrow 10^{-11}\ sec^{-1}$

H_2 の生成率

$\beta n(H_2) = R n_H^2$ $R = \frac{n(H_2)}{n_H^2}\beta$ $f \equiv \frac{2n(H_2)}{2n(H_2)+n_H} \simeq 2\frac{n(H_2)}{n_H} \simeq 10^{-17}\ cm^3\ s^{-1}$

$= \frac{\beta}{2n_H} f$

1) radiative association

$H + H \rightarrow H_2 + h\nu$ $A = 10^{-5}\ s^{-1}$ $k = 10^{-29}$

2) ion-molecule

$H + H^+ \xrightarrow{k_2} H_2^+ + h\nu$ $k_2 = 10^{-18}$

$H_2^+ + H \xrightarrow{k_3} H_2 + H$ $k_3 = 10^{-9}$ $R = 10^{-18}\left(\frac{n_{H^+}}{n}\right) cm^3\ s^{-1}$

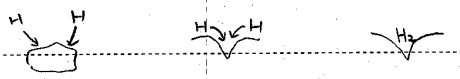
$H + H^- \xrightarrow{k_4} H_2 + e$ $k_4 = 10^{-9}$ $R = 10^{-9}\left(\frac{n_{H^-}}{n}\right)$

$n(H^-) = 10^{-9} n_H n_e$ $= 10^{-21}\ cm^3\ s^{-1}$

$n_e = 3\times10^{-4} n$

3) grain-surface reaction

吸着 → migrate → 反応 → jettison $H + H + grain \rightarrow H_2 + grain$



sticking

$k = \pi a^2 v P = 10^{-16} P\ cm^3\ s^{-1}$

$\pi a^2 m_g = 2.1\times10^{-21} n_H$

Star Formation $t_{ff} = \sqrt{\frac{3\pi}{32 G \rho_0}}$ Jun 28

OMC-1 BN 天体 IR星？

Becklin & Neugebauer (1968)

KL 天体

Klemman & Low

国 立 天 文 台

Galactic Center

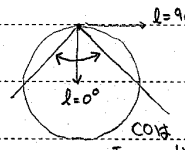
Jul 12

Visible Star $\lesssim$ 1 kpc < G.C 8.5 kpc　　Mass：星が大半

Radio Cont { Non thermal (Synchrotron) } Ionized Gas
{ Thermal (Free-Free) }

HI 21 cm ← Scale height 1 kpc

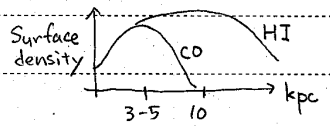
CO 2.6 μm ← 〃 60 pc



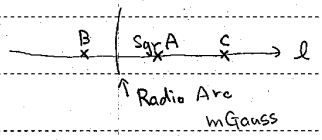
ゴダード
Mulit wavelength Universe

Inner disk に存在

星の観測 → 2.4 μm { $A_K / A_V \sim 1/10$
{ OH 夜光の窓



Radio Source



Sgr A Complex　Mini Spiral

Sgr A* = IRS 16 ?

Distance to the G.C.

8.0 ± 0.5 kpc
(IAU 8.5 kpc)

(1) H_2O Maser from Sgr B_2

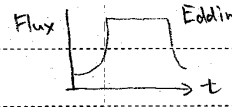
分泉が細い 10^{13} cm → V_{radial}　　$v = \Delta\theta R_0$ } => Spherical Expansion

T_B が高い 10^{15} K → VLBI → Proper Motion } => $R = 7.1 \pm 1.5$ kpc

(2) Globular Cluster 分布　8.5 ± 1.6 kpc

(3) RR Lyr Period-Luminosity Relation → 分布の中心 G.C.　$R = 8.1 \pm 0.4$ kpc

(4) X-ray Burst



$M_{NS} \sim 1.4 M_\odot$ → R = 6-7 kpc

国 立 天 文 台

AGN　QSO　$\sim 10^{-4}/\mathrm{Mpc}^3$　Jul 19

- $L_{QSO} \gtrsim 10^2 L_{MW}$
- Broad Line Regions　$\Delta V_{FWHM} \sim 5000\ \mathrm{km/s}$

(1) 明るい　$L \gtrsim 10^{46}\ \mathrm{ergs/s}$ → Eddington Limit

(2) Variable　$\Delta t \sim 1\ \mathrm{hours}$

$r < c\Delta t$

$\rho \sim \frac{10^8 M_\odot}{(10\mathrm{AU})^3} \sim 10^{21} M_\odot\ \mathrm{pc}^{-3}$　$10^{14}\ \mathrm{cm} \sim 10\mathrm{AU} \sim 3\times10^{-5}\ \mathrm{pc}$

Seyfert Galaxies　$\sim 0.1/\mathrm{Mpc}^3$ (c.f. normal $5/\mathrm{Mpc}^3$)

- $L_{sey} \sim 10 L_{MW}$
- 中心核

Type I ··· Strong Broad Lines

Type II ··· Narrow Line Region

Fueling

$L = \eta \dot{m} c^2$　$U = \frac{GMm}{r}$　$L = \frac{dU}{dt} = \frac{GM}{r}\dot{m}$　$r = 5R_g$ → $\eta = 0.1$

$L = 10^{46}\ \mathrm{ergs/s} \rightarrow \dot{m} \sim 2 M_\odot\ \mathrm{yr}^{-1}$

Accretion Disk

$L = \frac{GM\dot{m}}{2r} = 2\pi r^2 \sigma T^4$　$T = \left(\frac{3GM\dot{m}}{8\pi\sigma R_g}\right)\left(\frac{r}{R_g}\right)^{-3/4} = 6.3\times10^5 \left(\frac{\dot{m}}{\dot{m}_E}\right)^{1/4} M_8^{-\frac{1}{4}} \left(\frac{r}{r_g}\right)^{-\frac{3}{4}} \mathrm{K}$

$r = 5r_g$

$\dot{m} \sim \dot{m}_E$　$M \sim 10^8 M_\odot$ → 2×10^5 K, 150 Å

cf. $1 M_\odot$ → 2×10^7 K, 5 keV

Big Blue Bump?

(観測は困難)

Broad Line Region

$\Delta V = 5000\ \mathrm{km/s}$

Weak forbidden Lines　$n_e = 10^{10-11}\ \mathrm{cm}^{-3}$

Reverberation Mapping　$\tau \sim 2\text{-}22$ days

$R_{BLR} = 10$ Light days $\sim 3\times10^{16}\mathrm{cm}\ (10^{-2}\mathrm{pc})$

Narrow Line Region

$n_e \sim 10^3\ \mathrm{cm}^{-3}$

$R_{NLR} = 10\text{-}100\ \mathrm{pc}$

レポート　メ切 9/1 (水) 提出先 (1) 天文事務

枚数 A4 4枚程度

テーマ プロポーザルを書く.

講義で扱った内容に近い天文学的な問題を解決する研究プロポーザル { Observation / Theory / Simulation }

構成 1) 1/4 - 1/2 ページ Abstract

2) Introduction 今まで何が分かっているか？何が分かっていないか？

3) 本論 a) 何を明らかにするか

b) どのように明らかにするか

c) Feasibility 本当にできるか.

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